

Medical Research Paper Summarizer & Q&A Bot

Project submitted to the

SRM University – AP, Andhra Pradesh

for the partial fulfillment of the requirements to award the degree of

Bachelor of Technology

In

Computer Science and

Engineering School of

Engineering and Sciences

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Medical Research Paper Summariser & Q&A Bot

Project Description:

This project presents the design, implementation, and deployment of a cloud-native web application titled the Medical Research Paper Summariser & Q&A Bot. The system enables authenticated users to upload PDF documents containing medical research papers to a secure, cloud-hosted platform. Upon upload, the system generates a structured summary of the paper content and stores it persistently in a relational database. Users can subsequently retrieve the summary for any previously uploaded paper and, more importantly, engage in a natural language dialogue with the system to ask specific questions about the paper's content, methodology, findings, or implications. The question-and-answer functionality is accessible both through a RESTful API and through a conversational chatbot interface powered by Amazon Lex.

The entire system is built exclusively on Amazon Web Services, leveraging a carefully selected suite of managed and unmanaged services to achieve high availability, fault tolerance, horizontal scalability, and operational security. The infrastructure follows AWS Well-Architected Framework best practices across the pillars of operational excellence, security, reliability, performance efficiency, and cost optimization.

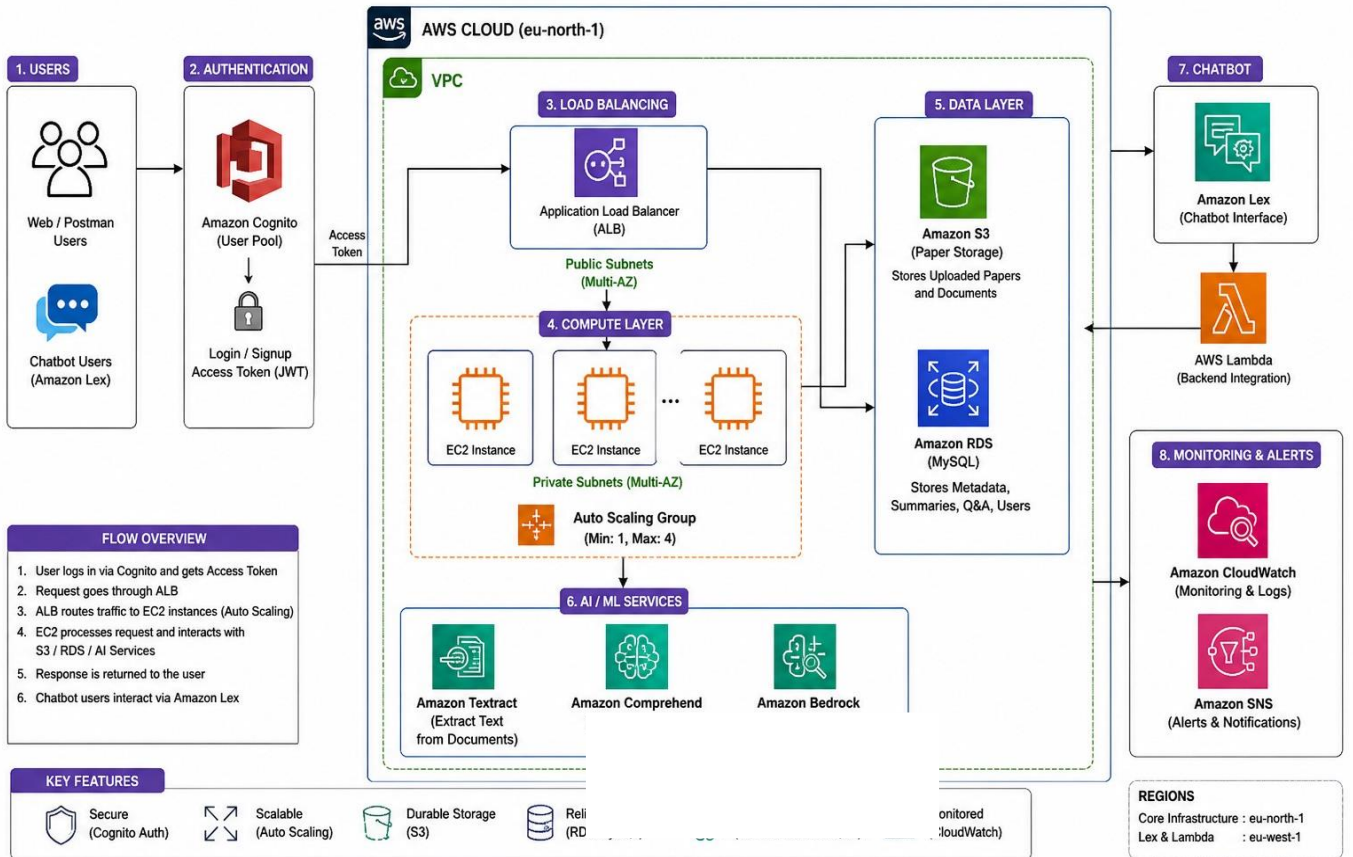
1.3 Domain Context

This project sits at the intersection of three domains: cloud computing, artificial intelligence, and biomedical informatics. Cloud computing provides the foundational infrastructure through which compute, storage, database, networking, and application services are delivered on demand with elastic scaling and a pay-per-use cost model. Artificial intelligence, specifically natural language processing, provides the algorithmic machinery for understanding, summarising, and answering questions about unstructured text. Biomedical informatics is the applied domain in which these capabilities are directed, addressing the information management challenges faced by medical professionals and researchers.

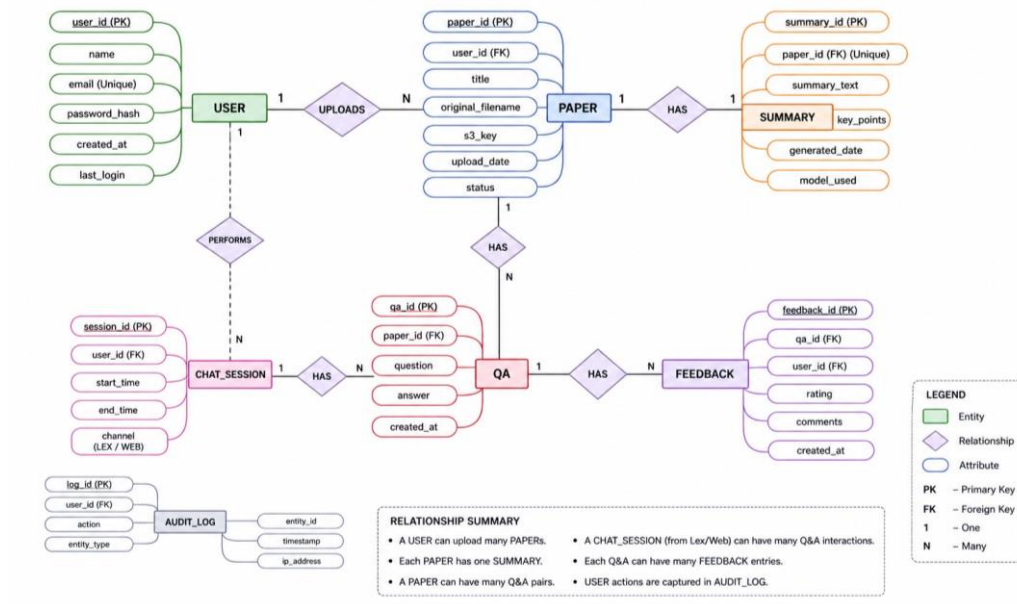
1.4 Scope of the Project

The scope of this project encompasses the complete software engineering lifecycle for the cloud-based system described above, including requirements analysis, architectural design, infrastructure provisioning, application development, integration of AI-driven NLP components, security configuration, monitoring setup, and functional testing. The project does not extend to the training or fine-tuning of custom language models; instead, it integrates with pre-trained models or NLP APIs as back-end processing components callable from the application layer.

AWS ARCHITECTURE



Entity Relationship (ER) Diagram:



Project WorkFlow:

1. AWS Account Setup and Login

Activity 1.1: Create AWS account

Activity 1.2: Log in to AWS Management Console

2. VPC and Networking Setup

Activity 2.1: Create VPC and subnets (public & private)

Activity 2.2: Configure Internet Gateway, NAT Gateway, and Route Tables

3. Security Configuration

Activity 3.1: Create Security Groups (ALB, App, Database)

Activity 3.2: Apply inbound/outbound access rules

4. Load Balancer Setup

Activity 4.1: Create Application Load Balancer

Activity 4.2: Configure Listener and Target Group

5. Compute Layer Setup

Activity 5.1: Launch EC2 instances in private subnets

Activity 5.2: Deploy Flask backend application

6. Data Layer Setup

Activity 6.1: Create Amazon S3 bucket for paper storage

Activity 6.2: Create Amazon RDS database

7. AI, Authentication & Chatbot

Activity 7.1: Integrate Textract & Comprehend for summarisation

Activity 7.2: Configure Cognito login and Lex chatbot

8. Monitoring, Testing & Deployment

Activity 8.1: Set up CloudWatch dashboard and SNS alerts

Activity 8.2: Test APIs, chatbot, and complete system flow

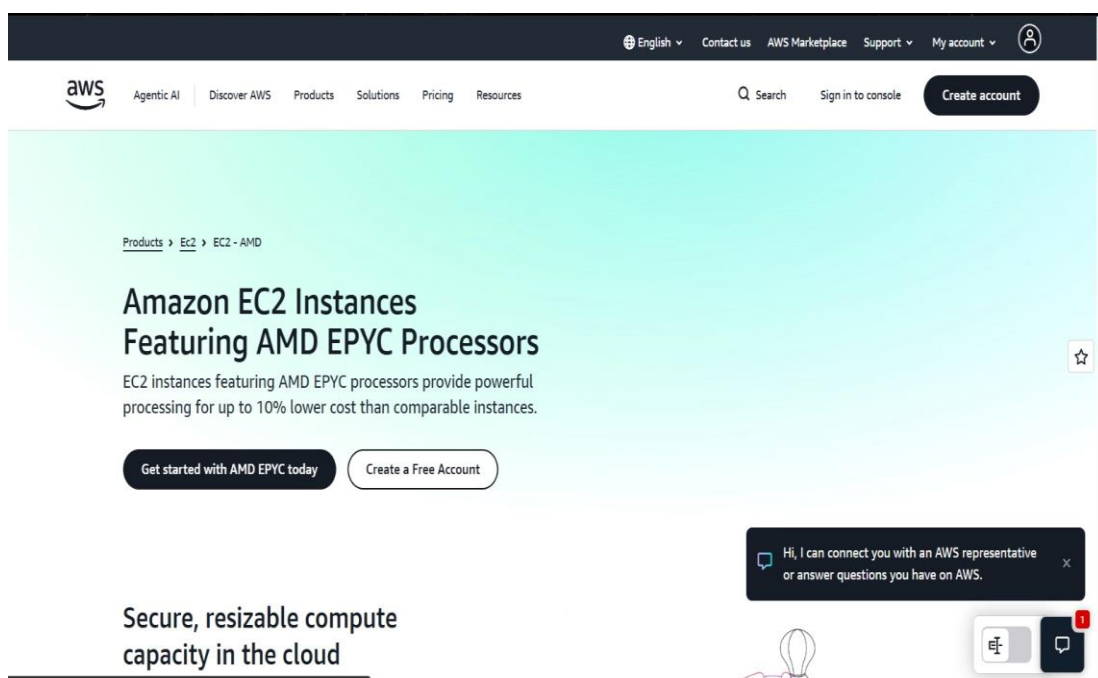
1. Testing and Deployment

Activity 8.1: Conduct functional testing to verify user registration, login, book requests, and notifications.

Milestone 1: AWS Account Setup and Login

Activity 1.1: Create AWS Account

- Sign up for an AWS account using a valid email.
- Configure billing and verification details.
- Enable security features like MFA.



Activity 1.2: Log in to AWS Management Console

- Log in using AWS credentials.
- Explore dashboard and services.
- Select deployment region.



Sign In

Access your AWS account by user type.

User type *(not sure?)*

☒ **Root user**

Account owner that performs tasks requiring unrestricted access.

☐ **IAM user**

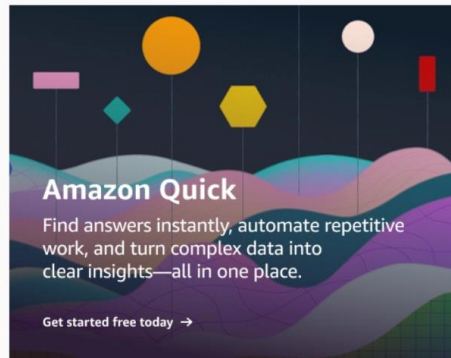
User within an account that performs daily tasks.

Email address

Next

OR

[New to AWS? Sign up](#)



Milestone 2: VPC and Networking Setup

Activity 2.1: Create VPC and Subnets

- Create a custom VPC (10.0.0.0/16).
- Create public and private subnets across multiple Availability Zones.
- Separate subnets for application and database layers.

The screenshot shows the AWS VPC dashboard for the Stockholm region. The left sidebar contains navigation links for VPCs, Subnets, Route tables, Internet gateways, Egress-only internet gateways, DHCP option sets, Elastic IPs, Managed prefix lists, NAT gateways, Peering connections, Route servers, Security, Network ACLs, Security groups, PrivateLink and Lattice, and Getting started. The main content area displays 'Resources by Region' for Stockholm, showing counts for VPCs (2), Subnets (9), Route Tables (5), Internet Gateways (2), Egress-only Internet Gateways (0), Endpoint Services (0), NAT Gateways (1), VPC Peering Connections (0), Network ACLs (2), and Security Groups (6). Buttons for 'See all regions' are provided for each resource type. The right sidebar includes 'Service Health', 'Settings' (Block Public Access, Zones, Console Experiments), 'Additional Information' (VPC Documentation, All VPC Resources, Forums, Report an Issue), and 'AWS Network Manager'.

The screenshot shows the AWS Subnets page for the Stockholm region. The left sidebar is identical to the VPC dashboard. The main content area displays a table of subnets with columns for Name, Subnet ID, State, VPC, Block Public..., and IPv4 CIDR. The table lists 9 subnets, all in an 'Available' state. Below the table is a 'Select a subnet' dropdown menu. The right sidebar includes 'Actions' and 'Create subnet' buttons, along with a 'Last updated' timestamp.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
private-data-1a	subnet-09ae501632f58d9c4	Available	vpc-0287973a4041bd680 vpc...	Off	10.0.21.0/24
-	subnet-0d30e77da0fa9a786	Available	vpc-08db690c6163c3993	Off	172.31.32.0/2
private-data-1b	subnet-030353fa27d1c1b2f	Available	vpc-0287973a4041bd680 vpc...	Off	10.0.22.0/24
public-sub-1a	subnet-0e91049311b194924	Available	vpc-0287973a4041bd680 vpc...	Off	10.0.1.0/24
private-app-1b	subnet-03990fe05ae19fc10	Available	vpc-0287973a4041bd680 vpc...	Off	10.0.12.0/24
private-app-1a	subnet-028dc214c49187be9	Available	vpc-0287973a4041bd680 vpc...	Off	10.0.11.0/24
public-sub-1b	subnet-001257b41a4bb18c7	Available	vpc-0287973a4041bd680 vpc...	Off	10.0.2.0/24

Activity 2.2: Configure Internet Gateway, NAT Gateway, and Route Tables

- Attach Internet Gateway to VPC for public access.
- Create NAT Gateway for private subnet internet access.
- Configure route tables for proper traffic routing.

The image displays two screenshots of the AWS Management Console, specifically the VPC dashboard, illustrating the configuration of network infrastructure.

Top Screenshot: Internet Gateways

The top screenshot shows the "Internet gateways (2)" page. The left sidebar lists navigation options under "Virtual private cloud", "Security", and "PrivateLink and Lattice". The main content area displays a table of existing Internet gateways:

Name	Internet gateway ID	State	VPC ID	Owner
igw-medical-app	igw-067b0bb76da780d52	Attached	vpc-0287973a4041bd680 vpc-medica...	235063677572
-	igw-07e1c84612ee9e8f0	Attached	vpc-08db690c6163c3993	235063677572

Below the table, there is a section titled "Select an internet gateway above" with a search bar and a dropdown menu.

Bottom Screenshot: NAT Gateways

The bottom screenshot shows the "NAT gateways (1)" page. The left sidebar is identical to the top screenshot. The main content area displays a table of existing NAT gateways:

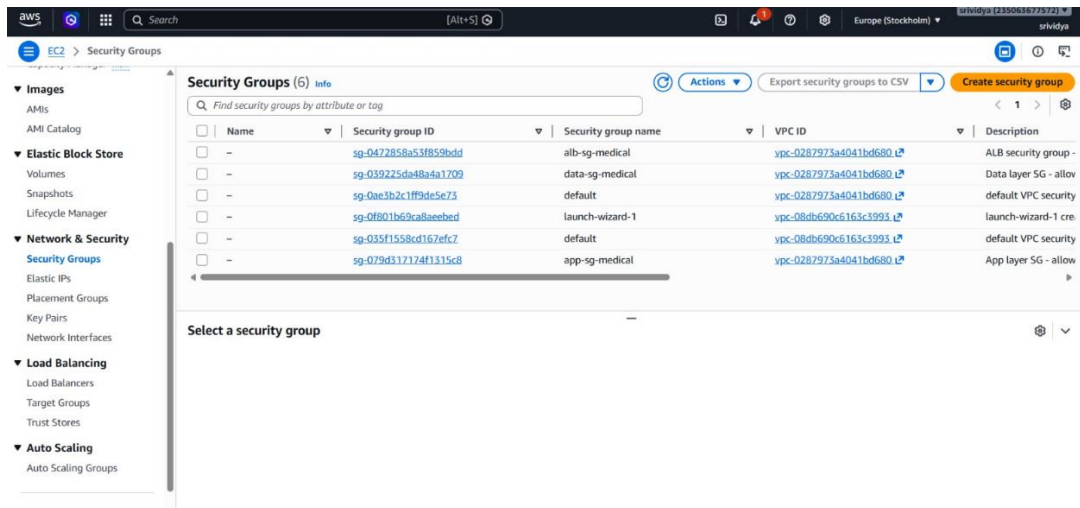
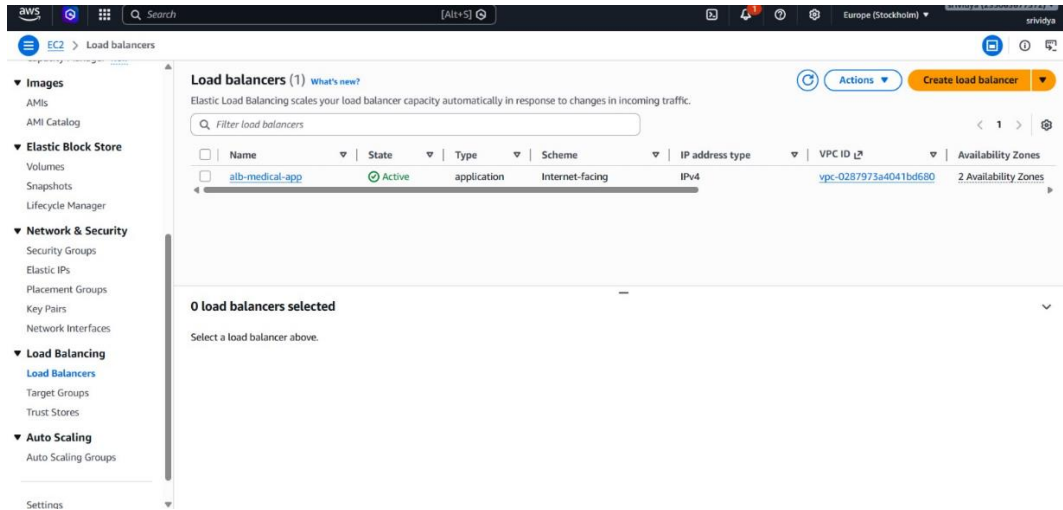
Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID	P
nat-gw-medical	nat-04a668a1564500f27	Public	Available	-	Zonal	-	1

Below the table, there is a section titled "Select a NAT gateway" with a search bar and a dropdown menu.

Milestone 3: Security Configuration

Activity 3.1: Create Security Groups

- Create security groups for ALB, EC2 instances, and RDS database.
- Define access rules between components.



Activity 3.2: Configure Access Rules

- Allow HTTP/HTTPS traffic for ALB.
- Allow application traffic (port 8080) from ALB to EC2.
- Allow database access (port 5432) from EC2 only.

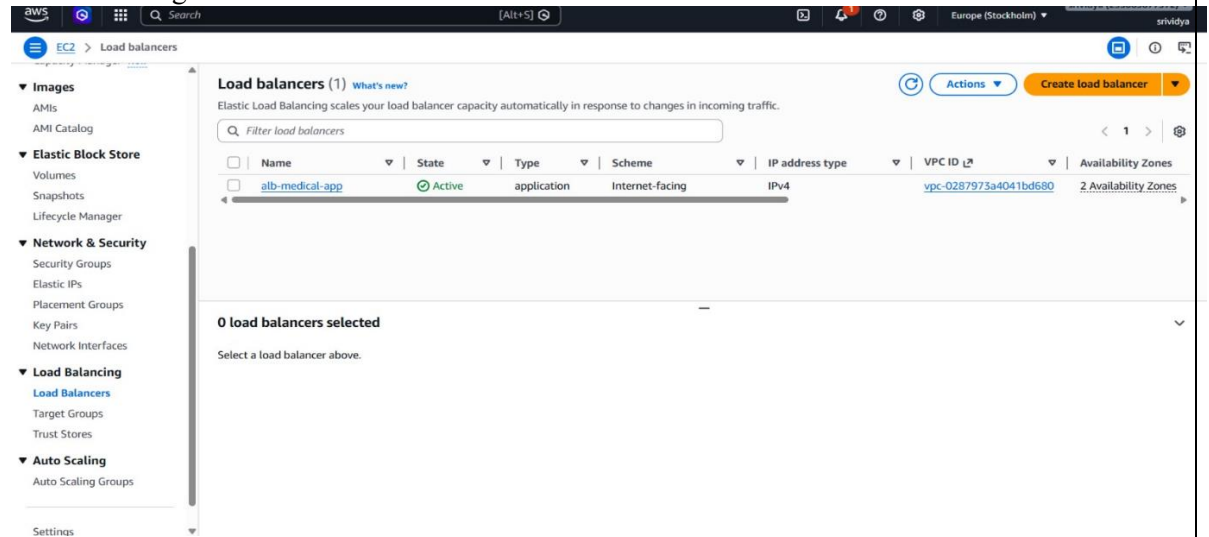
The screenshot shows the AWS Management Console interface for an EC2 instance. The left sidebar contains navigation links for EC2, Instances, and various AWS services. The main content area displays the 'Instance summary for i-074377bebcaba79c0 (app-server-1)'. The instance is in a 'Running' state. The summary includes the following details:

- Instance ID:** i-074377bebcaba79c0
- IPV6 address:** -
- Hostname type:** IP name: ip-10-0-11-138.eu-north-1.compute.internal
- Answer private resource DNS name:** -
- Auto-assigned IP address:** -
- IAM role:** EC2-SSM-Medical-App-Role
- IMDSv2:** Required
- Operator:** -
- Public IPv4 address:** -
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-10-0-11-138.eu-north-1.compute.internal
- Instance type:** t3.micro
- VPC ID:** vpc-0287973a4041bd680 (vpc-medical-app)
- Subnet ID:** subnet-028dc214c49187be9 (private-app-1a)
- Instance ARN:** arn:aws:ec2:eu-north-1:235063677572:instance/i-074377bebcaba79c0
- Private IPv4 addresses:** 10.0.11.138
- Public DNS:** -
- Elastic IP addresses:** -
- AWS Compute Optimizer finding:** Opt-in to AWS Compute Optimizer for recommendations. | Learn more
- Auto Scaling Group name:** -
- Managed:** false

Milestone 4: Load Balancer Setup

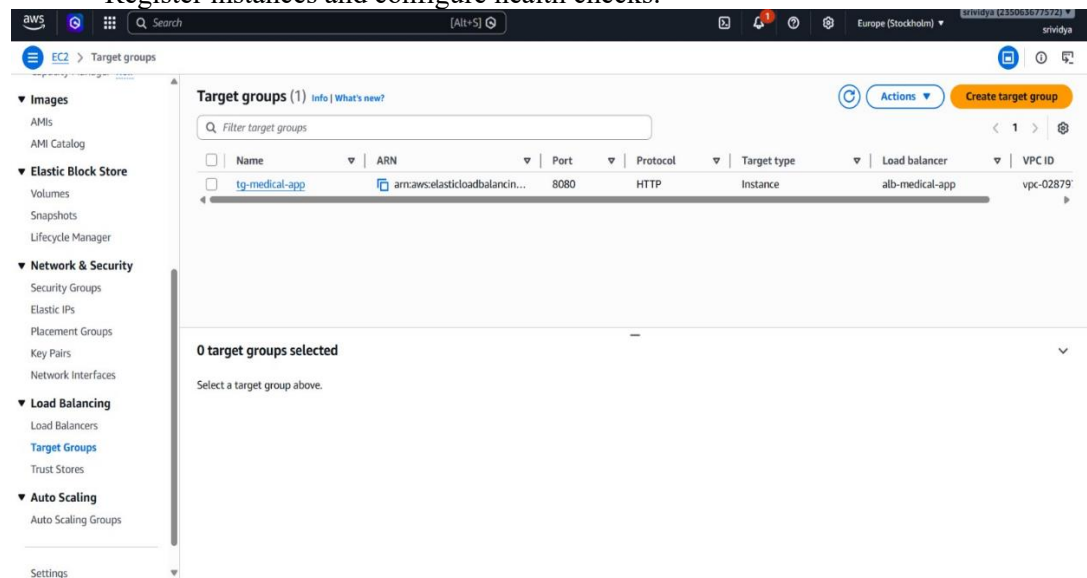
Activity 4.1: Create Application Load Balancer

- Create ALB in public subnets.
- Configure listeners for HTTP traffic.



Activity 4.2: Configure Target Group

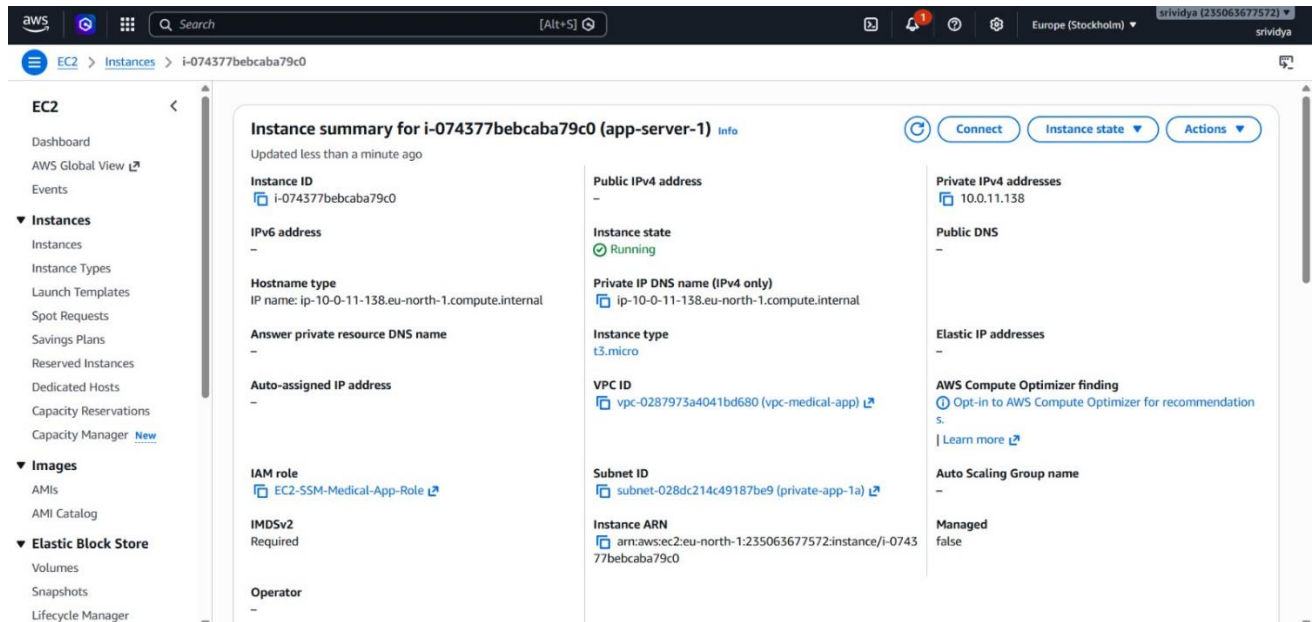
- Create target group for EC2 instances.
- Register instances and configure health checks.



Milestone 5: Compute Layer Setup

Activity 5.1: Launch EC2 Instances

- Launch EC2 instances in private subnets.
- Attach IAM role and security groups.
- Configure instance settings.



Activity 5.2: Deploy Backend Application

- Install required software (Python, Flask).
- Deploy Medical Research Paper Summariser application.
- Start application on port 8080.

Session ID: root-9frxpqbt9grajncdvsldvbh3je

[Shortcuts](#)

Instance ID: i-074377bebcaba79c0 (app-server-1)

[Terminate session](#)

```
sh-5.2$ cd /opt/medical-app
kill -9 python
python3 app.py
kill: killing pid 69979 failed: Operation not permitted
* Serving Flask app 'app'
* Debug mode: off
Address already in use
Port 8080 is in use by another program. Either identify and stop that program, or start the server with a different port.
sh-5.2$
```

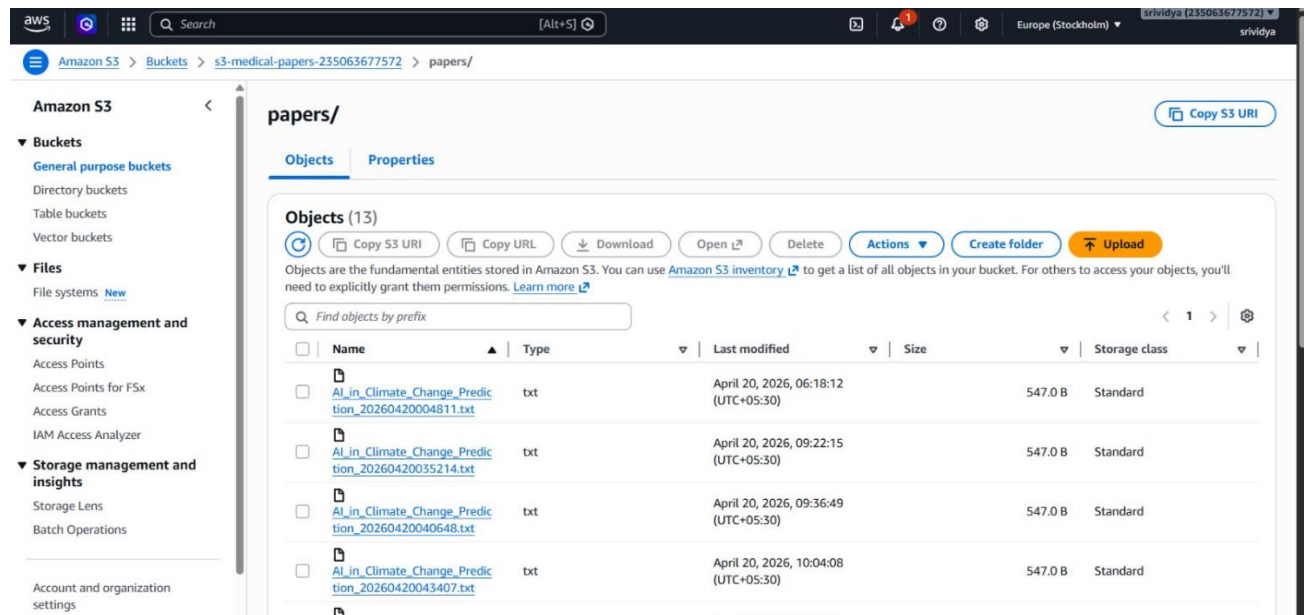
Pretty-print ☐

```
{"database": "connected", "s3": "connected", "status": "healthy", "time": "2026-05-04 16:29:15.568176"}
```

Milestone 6: Data Layer Setup

Activity 6.1: Create Amazon S3 Bucket

- Create S3 bucket for storing research papers.
- Configure permissions and access policies.

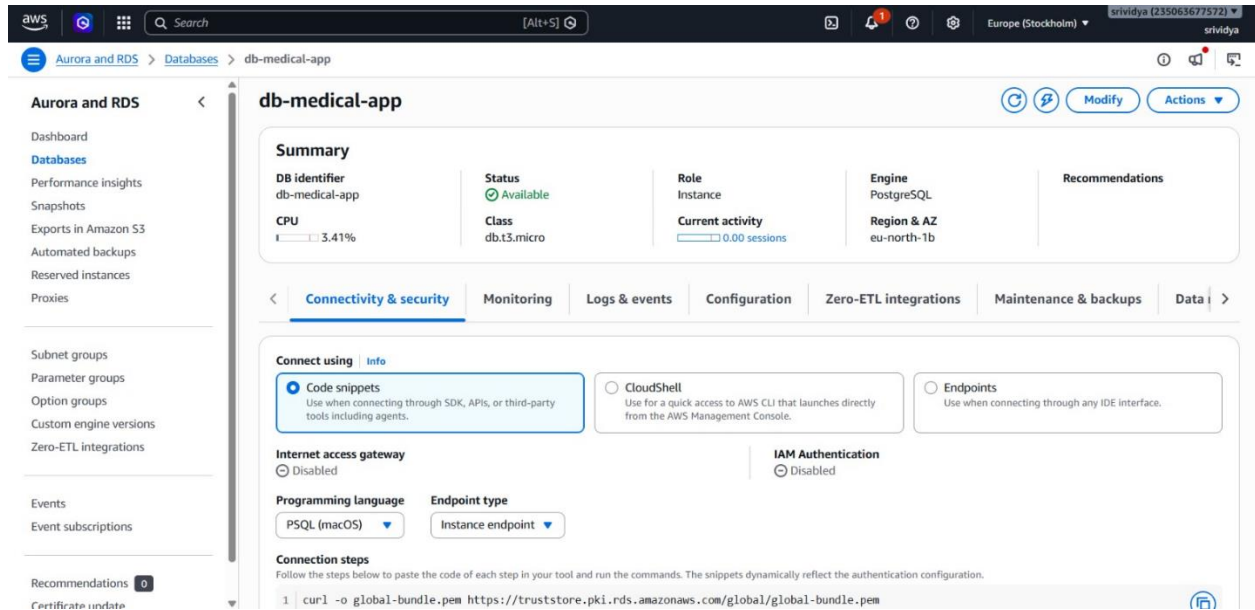


The screenshot displays the Amazon S3 console interface. The left sidebar shows the navigation menu with categories like Buckets, Files, Access management and security, and Storage management and insights. The main content area shows the 'papers/' bucket with a list of 13 objects. The objects are text files with names starting with 'AI_in_Climate_Change_Prediction_'. The table below summarizes the objects shown in the screenshot.

Name	Type	Last modified	Size	Storage class
AI_in_Climate_Change_Prediction_20260420004811.txt	txt	April 20, 2026, 06:18:12 (UTC+05:30)	547.0 B	Standard
AI_in_Climate_Change_Prediction_20260420035214.txt	txt	April 20, 2026, 09:22:15 (UTC+05:30)	547.0 B	Standard
AI_in_Climate_Change_Prediction_20260420040648.txt	txt	April 20, 2026, 09:36:49 (UTC+05:30)	547.0 B	Standard
AI_in_Climate_Change_Prediction_20260420043407.txt	txt	April 20, 2026, 10:04:08 (UTC+05:30)	547.0 B	Standard

Activity 6.2: Create Amazon RDS Database

- Create PostgreSQL database in private subnet.
- Configure database security group.
- Connect application to database.



Milestone 7: AI, Authentication & Chatbot Integration

Activity 7.1: Integrate Textract and Comprehend

- Use Textract to extract text from uploaded documents.
- Use Comprehend to generate summaries.

Medical Research AI Assistant

Upload Paper

Upload Paper

Get Summary

13

Generate Summary

["id":13,"summary":"Artificial Intelligence (AI) is transforming healthcare by improving diagnosis accuracy and patient monitoring. Machine learning models analyze large medical data to detect diseases early. AI systems assist doctors in identifying patterns and improving treatment decisions. Possible advantages: challenges such as data privacy and

Ask Question

Get Answer

Activity 7.2: Configure Cognito and Lex

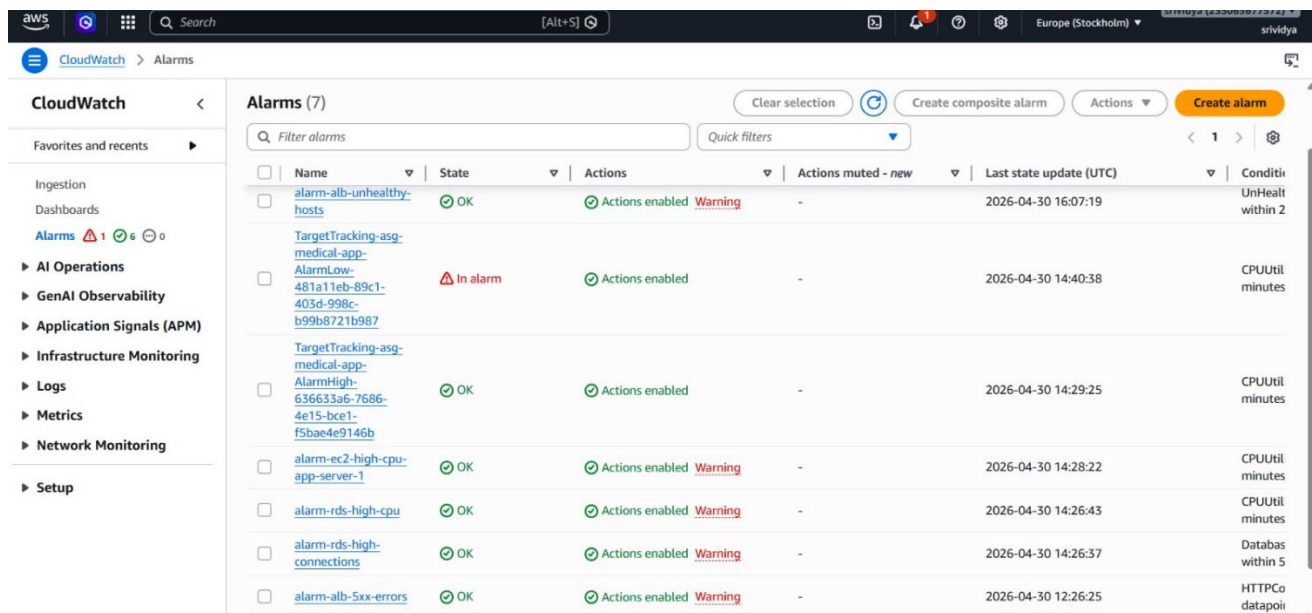
- Set up Cognito for user authentication.
- Create Lex chatbot for Q&A interaction.
- Integrate chatbot with backend using Lambda.

The screenshot displays the Amazon Lex console interface. The top navigation bar includes the AWS logo, a search bar, and the user's profile (srividya). The main content area shows the configuration for a bot named "medical-research-bot" in the "English (US)" language. The "Draft version" is selected, and a "Successfully built" status is visible. The "Conversation structure" section shows 2 intents and 0 slot types. The "Language details" section shows the voice as "Danielle". A "Test Draft version" window is open on the right, showing a test message: "What is the conclusion of paper 4". The test results show a response from the bot: "Paper 'Smart Urban Farming Using Vertical Hydroponics' discusses: Urban areas are facing increasing challenges in food production due to limited land availability and growing populations. Vertical hydroponic farming offers an innovative solution by allowing crops to be grown in stacked layers using nutrient-rich water instead of soil. This method significantly". The test status is "Ready for complete testing".

Milestone 8: Monitoring, Testing & Deployment

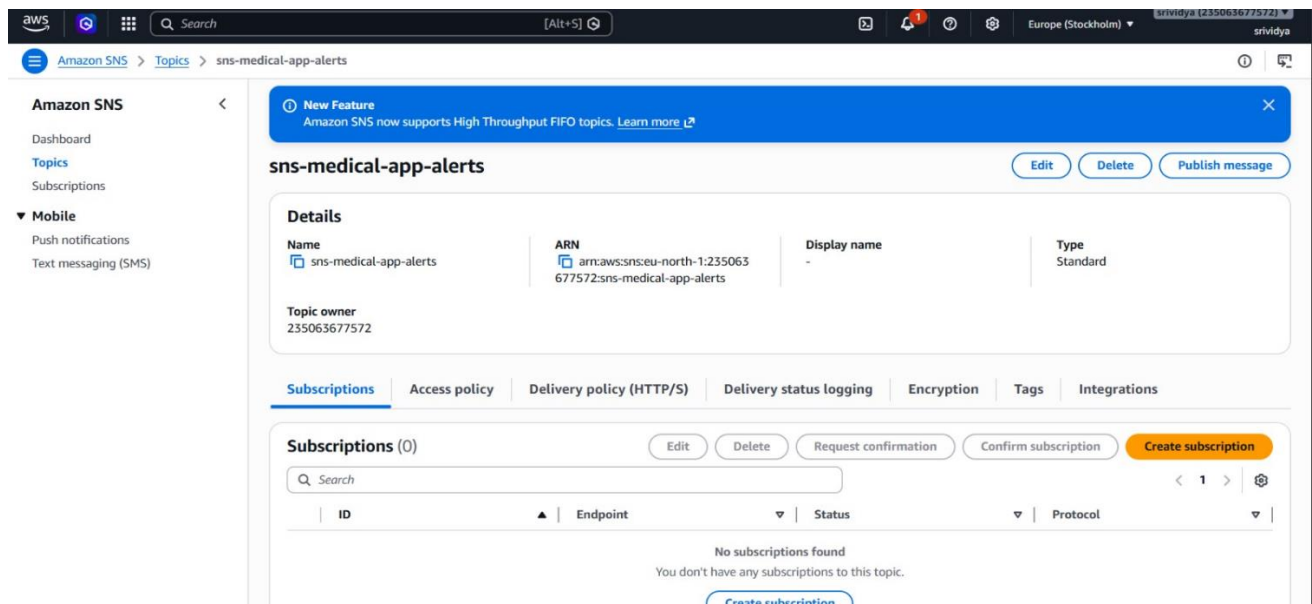
Activity 8.1: Set up CloudWatch Dashboard and SNS Alerts

- Create CloudWatch dashboard for monitoring system performance.
- Configure alarms for EC2, ALB, and RDS metrics.
- Create SNS topic and subscribe users for alerts.



The screenshot shows the AWS CloudWatch Alarms console. The left sidebar contains navigation links for CloudWatch, Alarms, Favorites and recents, Ingestion, Dashboards, and a list of monitoring categories including AI Operations, GenAI Observability, Application Signals (APM), Infrastructure Monitoring, Logs, Metrics, Network Monitoring, and Setup. The main area displays a table of 7 alarms. The table columns are Name, State, Actions, Actions muted - new, Last state update (UTC), and Condition. The alarms listed are:

Name	State	Actions	Actions muted - new	Last state update (UTC)	Condition
alarm-alb-unhealthy-hosts	OK	Actions enabled	Warning	2026-04-30 16:07:19	UnHealthy within 2
TargetTracking-asg-medical-app-AlarmLow-481a11eb-89c1-403d-998c-b99b8721b987	In alarm	Actions enabled	-	2026-04-30 14:40:38	CPUUtil minutes
TargetTracking-asg-medical-app-AlarmHigh-636633a6-7686-4e15-bce1-f5bae4e9146b	OK	Actions enabled	-	2026-04-30 14:29:25	CPUUtil minutes
alarm-ec2-high-cpu-app-server-1	OK	Actions enabled	Warning	2026-04-30 14:28:22	CPUUtil minutes
alarm-rds-high-cpu	OK	Actions enabled	Warning	2026-04-30 14:26:43	CPUUtil minutes
alarm-rds-high-connections	OK	Actions enabled	Warning	2026-04-30 14:26:37	Databases within 5
alarm-alb-5xx-errors	OK	Actions enabled	Warning	2026-04-30 12:26:25	HTTPCo datapoi



The screenshot shows the Amazon SNS console for the topic 'sns-medical-app-alerts'. The left sidebar contains navigation links for Amazon SNS, Topics, Subscriptions, and Mobile. The main area displays the topic details, including the Name, ARN, Display name, and Type. Below the details, there are tabs for Subscriptions, Access policy, Delivery policy (HTTP/S), Delivery status logging, Encryption, Tags, and Integrations. The Subscriptions tab is active, showing a list of subscriptions (0) with columns for ID, Endpoint, Status, and Protocol. A message at the bottom states: 'No subscriptions found. You don't have any subscriptions to this topic.' A 'Create subscription' button is visible at the bottom.

Activity 8.2: Test APIs and Complete System Flow

- Test API endpoints such as upload, summary, and Q&A.
- Verify chatbot functionality.
- Validate end-to-end workflow from upload to response.

Conclusion:

This project has successfully designed, implemented, deployed, and tested a complete cloud-native system for medical research paper summarisation and question answering on Amazon Web Services. The system fulfils all stated objectives: it provides secure, authenticated access through Amazon Cognito; scales elastically under variable load through an Auto Scaling Group managed by target tracking policies; stores uploaded papers durably in Amazon S3 with server-side encryption; persists metadata and summaries in Amazon RDS PostgreSQL; provides a RESTful API with four well-defined endpoints for health checking, uploading, summarising, and querying papers; and offers a conversational chatbot interface through Amazon Lex and AWS Lambda.

The architectural design demonstrates the practical application of AWS Well-Architected Framework principles. The multi-AZ VPC with separate public, private application, and private data subnet tiers enforces a clear network security boundary. The removal of SSH access in favour of SSM Session Manager eliminates a common attack vector. Least-privilege security group rules ensure that each component can communicate only with the specific services it requires. VPC Flow Logs, CloudWatch metrics, and automated alarms provide comprehensive operational visibility.

From a technical perspective, this project illustrates the power of cloud-managed services to reduce the operational burden of building and running a complex, multi-component system. Services such as Cognito, Lex, Lambda, RDS, and CloudWatch each handle substantial underlying complexity, including scaling, patching, high availability, and security, that would require significant dedicated engineering effort if implemented on-premises or with unmanaged infrastructure. By composing these services according to established cloud design patterns, a small development team can build and operate a system of considerable sophistication.

The limitations identified, including the HTTP-only ALB listener, single-region deployment, and synchronous NLP processing, represent known technical debt items that would be addressed before any production deployment. The future enhancements described in the preceding chapter provide a clear roadmap for the continued evolution of the system toward a robust, production-grade tool.

In conclusion, the Medical Research Paper Summariser & Q&A Bot project demonstrates that cloud computing, when applied thoughtfully and in accordance with established best practices, is an effective platform for delivering intelligent, scalable, and secure applications in the healthcare and biomedical informatics domain. The complete, tested, and documented system produced by this project stands as evidence of the team's proficiency in cloud architecture, AWS service integration, backend development, and AI-assisted application design. complete, tested, and documented system produced by this project stands as evidence of the team's proficiency in cloud architecture, AWS service integration, backend development, and AI-assisted application design.